Using tidymodels in forest modeling workflows



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11 November 2026
SOMENS/NEMO Meeting

Workshop resources

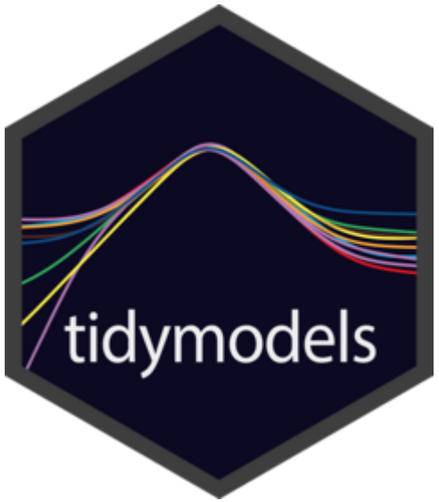
- Welcome!
- Access the code, data and slides from https://github.com/mbrussell/tidymodels/
- We'll conclude by 3:00 p.m.



Objectives

1. Understand the core packages and functions within tidymodels and how they can be used for common forest modeling scenarios.
2. Be able to design workflows that allow for training and testing models to evaluate and validate their performance.
3. Understand how to work with model output in an efficient manner so that results can be easily interpreted and visualized.

tidymodels




```
install.packages("tidymodels")
library(tidymodels)
```

- A collection of R packages for modeling (e.g., regression, machine learning)
- Uses tidyverse principles (good for data wrangling and visualization)
- Sets up workflows that are reproducible

tidymodels is a collection of many packages

There are ~8 core packages in `tidymodels`. We'll focus on:

- `rsample` - Splits data into training and testing sets
- `recipes` - Preprocesses data
- `parsnip` - Specifies models
- `broom` - Converts model objects into tidy datasets
- `yardstick` - Assesses model performance

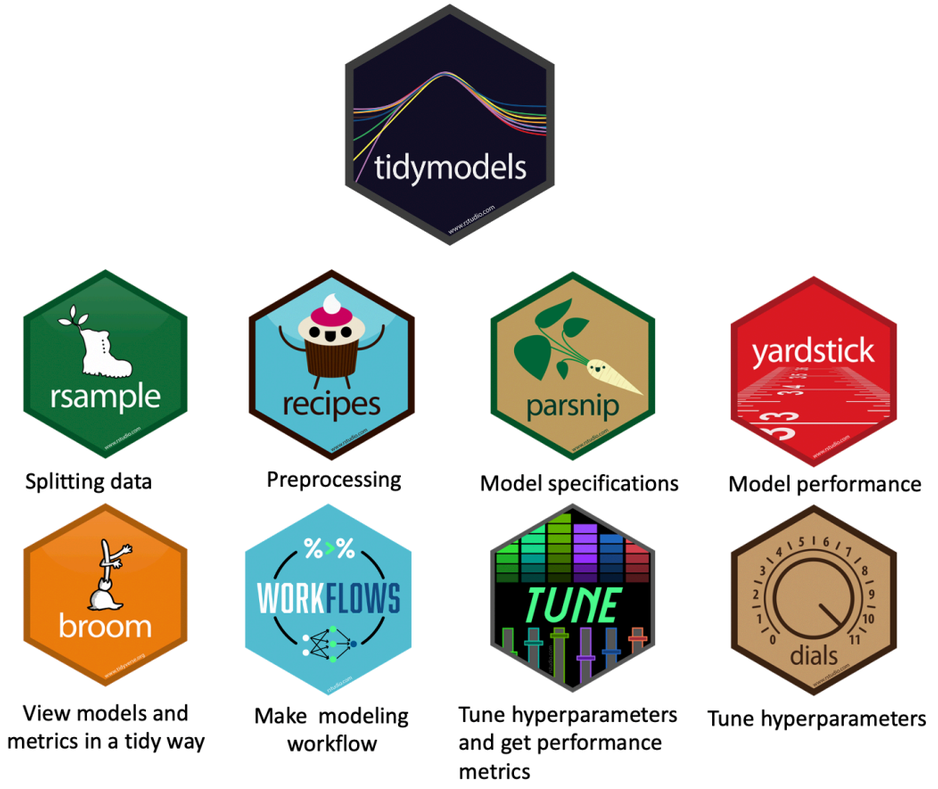


Image: rpubs.com/chenx/tidymodels_tutorial

Our dataset

- NSVB_TREE.csv
- Tree measurements of volume, weight, and physical properties (*n* = 12,000+)
- Biomass components for stem wood, bark, branches, and foliage
- Available from Virginia Tech's Data Repository



Goal: Predict aboveground tree biomass for various species groups

- There are 114 different species in the dataset
- We will group species into one of 10 Jenkins species groups.

```
tree |>
  group_by(COMMON_NAME) |>
  summarize(`Num trees` = n(),
            `Mean DBH (cm)` = round(mean(dbh_cm, na.r
            `Min DBH (cm)` = min(dbh_cm, na.rm=T),
            `Max DBH (cm)` = max(dbh_cm, na.rm=T),
            `Mean weight (kg)`= round(mean(bm_kg, na.
            `Min weight (kg)` = min(bm_kg, na.rm=T),
            `Max weight (kg)` = max(bm_kg, na.rm=T))
arrange(desc(`Num trees`))
```

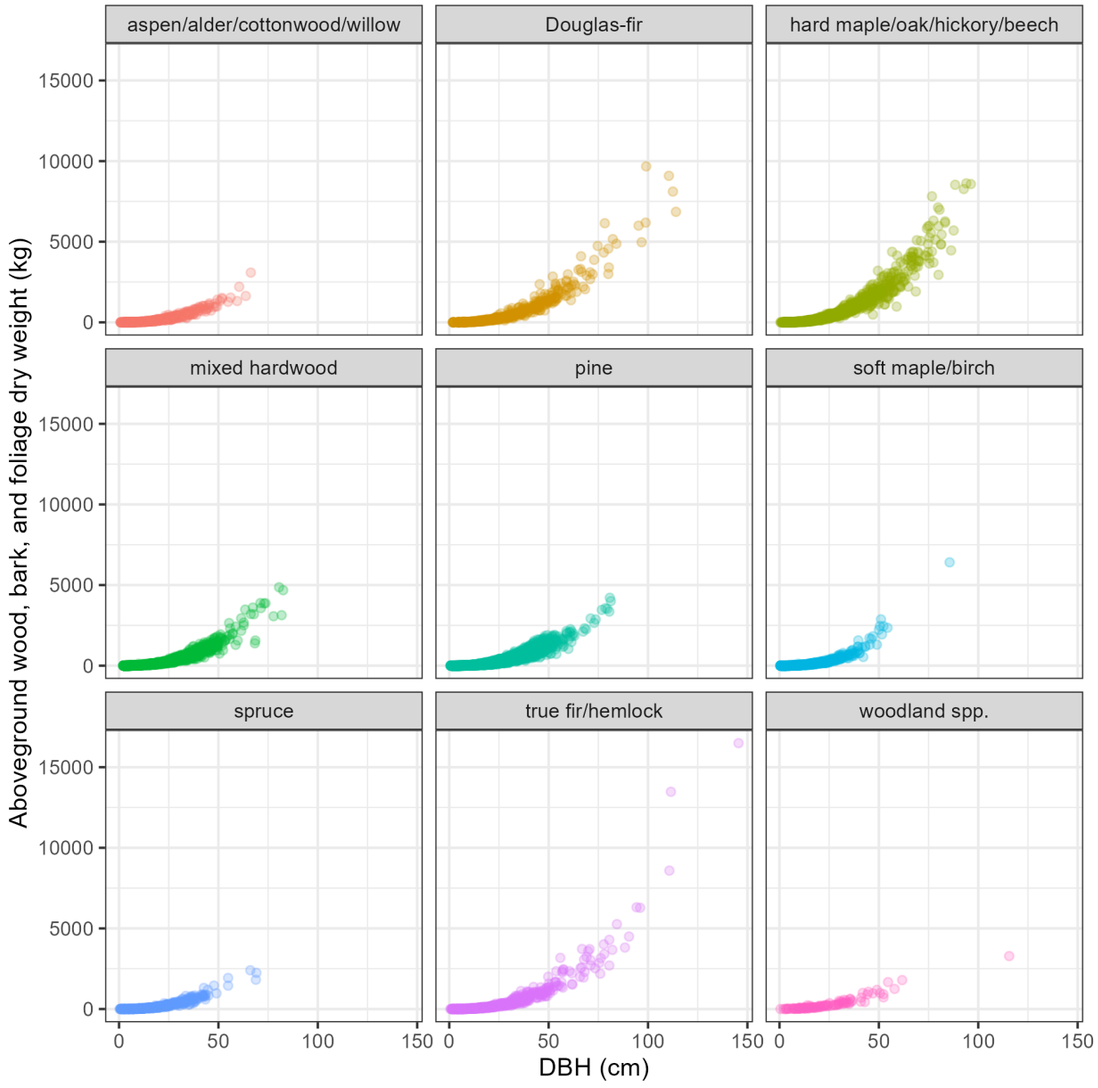
	COMMON_NAME	`Num trees`	`Mean DBH (cm)`	`Min DBH (cm)`	`Max DBH (cm)`	`Mean weight (kg)`	`Min weight (kg)`	`Max weight (kg)`
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	loblolly pine	2133	20.3	0.508	61.0	226.	0.231	2064.
2	slash pine	968	19.1	1.27	53.3	243.	0.844	1862.
3	sweetgum	715	17	2.29	82.6	224.	0.513	4682.
4	longleaf pine	586	25	3.05	52.3	437	0.848	1891.
5	Douglas-fir	494	29.3	1.78	114.	714.	0.408	9672.
6	shortleaf pine	449	22.9	2.54	50.0	308.	0.757	1625.
7	ponderosa pine	372	29.4	0.762	80.8	448.	0.263	4212.
8	red maple	363	14.2	0.508	85.6	161.	0.0771	6410.
9	white oak	303	23.7	2.54	87.6	559.	1.29	5997.
10	quaking aspen	273	19.7	0.762	60.5	226.	0.0408	2204.

i 104 more rows
i Use `print(n = ...)` to see more rows

Task 1: Explore the biomass dataset

⏱ 15 min

- Our primary variable of interest is `TT_DW_ADJ` (Aboveground wood, bark, and foliage dry weight)
- See the NSVB Data Dictionary for more information on the variables in the dataset.
- Head to `tidymodels_notebook.qmd` to explore to get started.



The modeling

Exercise	Functions
Refit a single species group	`workflow()`, `fit()`, `tidy()`
Refit to all species groups	`nest()`, `map()`
Compare fits to original Jenkins equation	Joining and interpreting coefficients
Predict biomass on new sites	Back-transformation

The Jenkins equation form

$$\ln(bm) = \beta_0 + \beta_1 \ln(dbh)$$

or equivalently, on the original scale:

$$bm = e^{\beta_0 + \beta_1 \ln(dbh)} = e^{\beta_0} \cdot dbh^{\beta_1}$$

- bm = total aboveground dry biomass (kg)
- dbh = diameter at breast height (cm)
- β_0, β_1 = coefficients, **one pair per species group**

- Linear on the log-log scale
- We can use `linear_reg()`

`broom`: turns model objects into tidy datasets

- Model objects from base R (e.g., `lm()`) store results in nested lists. These are awkward to work with.
- `broom` converts model onject output into **tibbles**, so you can use the tidyverse.
- Three key functions:
  - `tidy()` - Extracts model coefficients and related statistics
  - `glance()` - Provides overall model fit statistics
  - `augment()` - Adds predictions and residuals to the original data

Task 2: Refit the Jenkins et al. 2003 equation

⏱ 20 min

- Fit `ln(biomass) ~ ln(DBH)` separately for each Jenkins species group
- Compare your coefficients to the published values in `REF_SPECIES`
- Predict biomass on held-out sites and calculate bias in kg

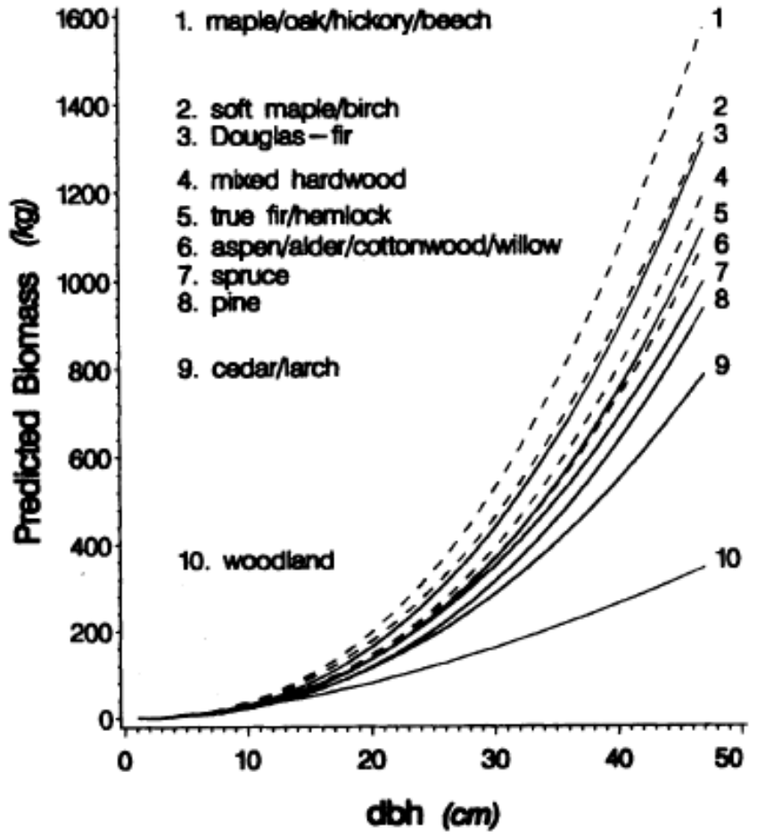
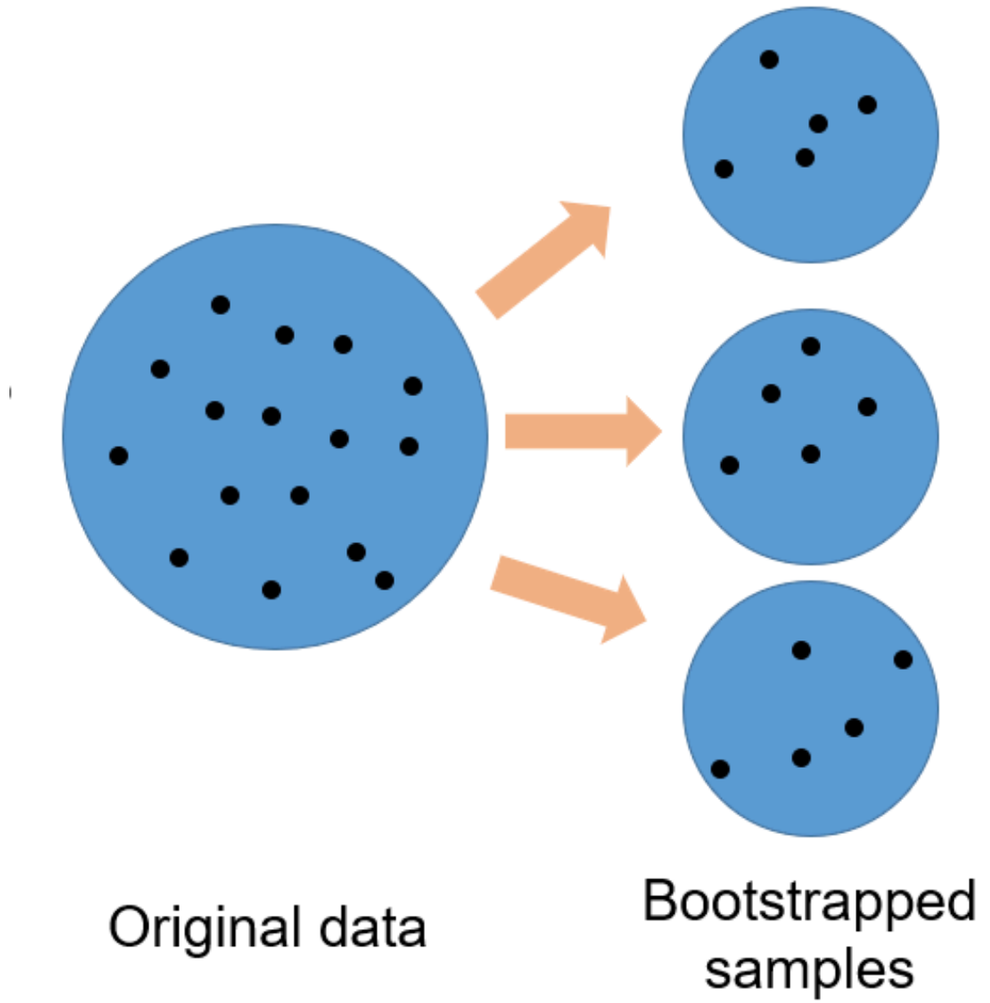


Figure 1. Graphs of ten equations for predicting total aboveground biomass by species group. Hardwoods are represented by dashed lines, softwoods by solid lines.

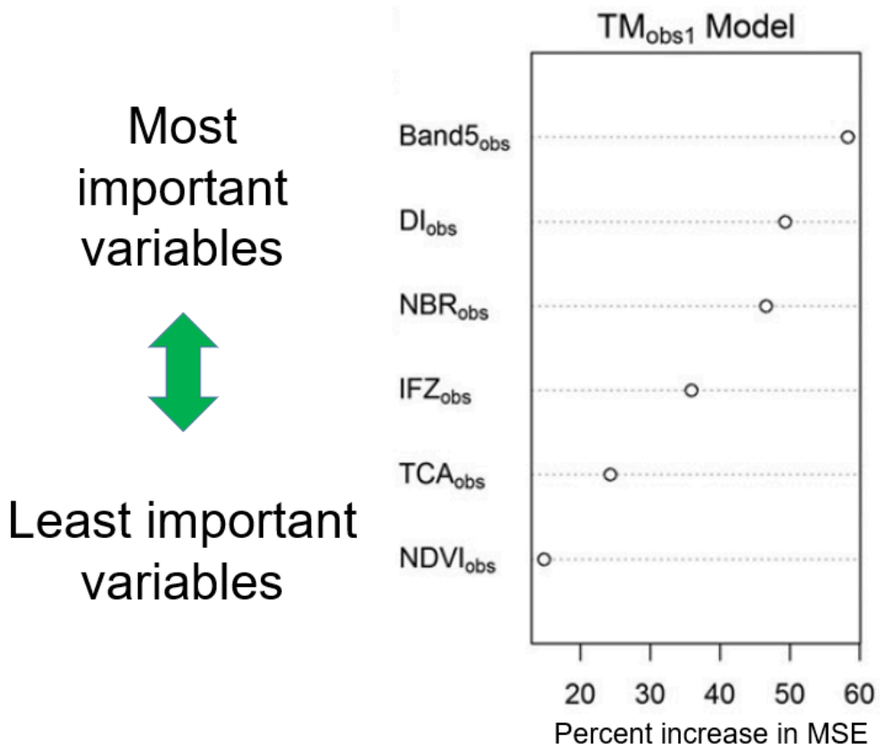
Random forests

- First, take several small samples of the data
- Any observations that aren't taken are called "out-of-bag" samples
- Build a set of regression trees on the sampled data and calculate mean square error



Variable importance in random forests

- If the percent increase in MSE after removing the variable is large, it's an important variable!
- If the percent increase in MSE is small, it's less important



Key arguments in `rand_forest()`

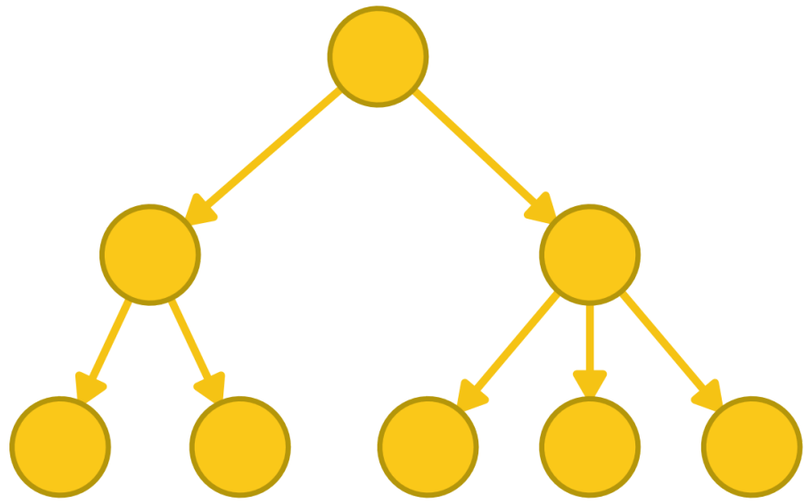
Argument	Meaning	Value
`trees`	Number of trees in the forest	e.g., 500
`mtry`	Predictors considered at each split	1 or 2 (DBH or species group)
`min_n`	Minimum observations to split a node	e.g., 5

```
rand_forest(
  trees = 500,
  mtry  = 2,
  min_n = 5) |>
  set_engine("ranger",
             importance = "permutation") |>
  set_mode("regression")
```

Task 3: Random forests to predict tree biomass

⏱ 15 min

- Specify a random forest with `rand_forest()` and the `ranger` engine
- Fit it to the same training data and predictors as the Jenkins refit
- Evaluate the model on held-out sites



How to compare models?

Metric	Measures
RMSE (kg)	Typical error
MAE (kg)	Mean absolute error
Bias (%)	Systematic over- or under-prediction

- Compute metrics on the original (kg) scale
- Use the held-out (test) data

The `yardstick` package

- Provides metric functions, e.g., `rmse()`, `mae()`, `rsq()`
- `metric_set()` bundles all metrics using a single function
- Outputs a dataset (i.e., a "tidy tibble")




```
# Get one metric
rmse(test_pred,
     truth = bm_kg,
     estimate = rf_kg)

# Get a bundle of metrics
biomass_metrics <- metric_set(rmse, mae, bias)
```

Task 4: Compare Jenkins refit to random forests

⏱ 15 min

- Combine predictions from the Jenkins refit and the random forests
- Calculate bias and error by Jenkins group with `yardstick` package
- Visualize where each model over- or under-predicts

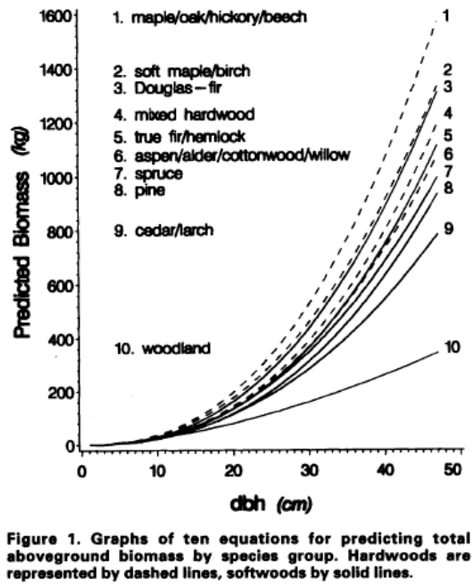
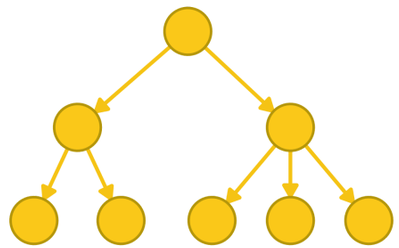


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Sidequests!

- See `tidymodels_notebook.qmd` for additional exercises to tackle with the data.



More `tidymodels` resources

- tidymodels.org
- *Tidy Modeling with R* (book)
- A few Arbor Analytics blog posts on tidymodels:
  - Predicting tree biomass for many species
  - Bootstrapped resampling to model tree biomass